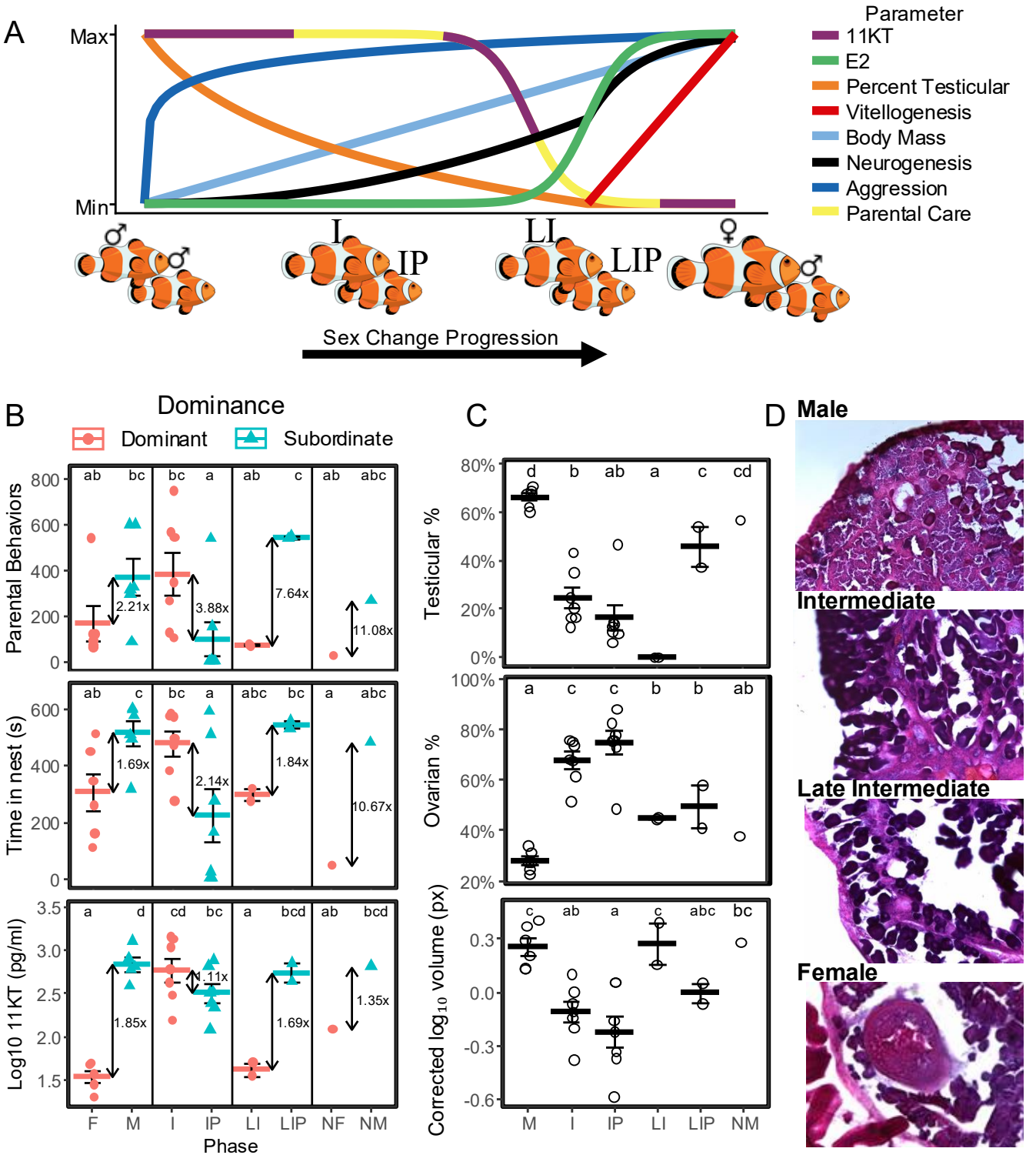
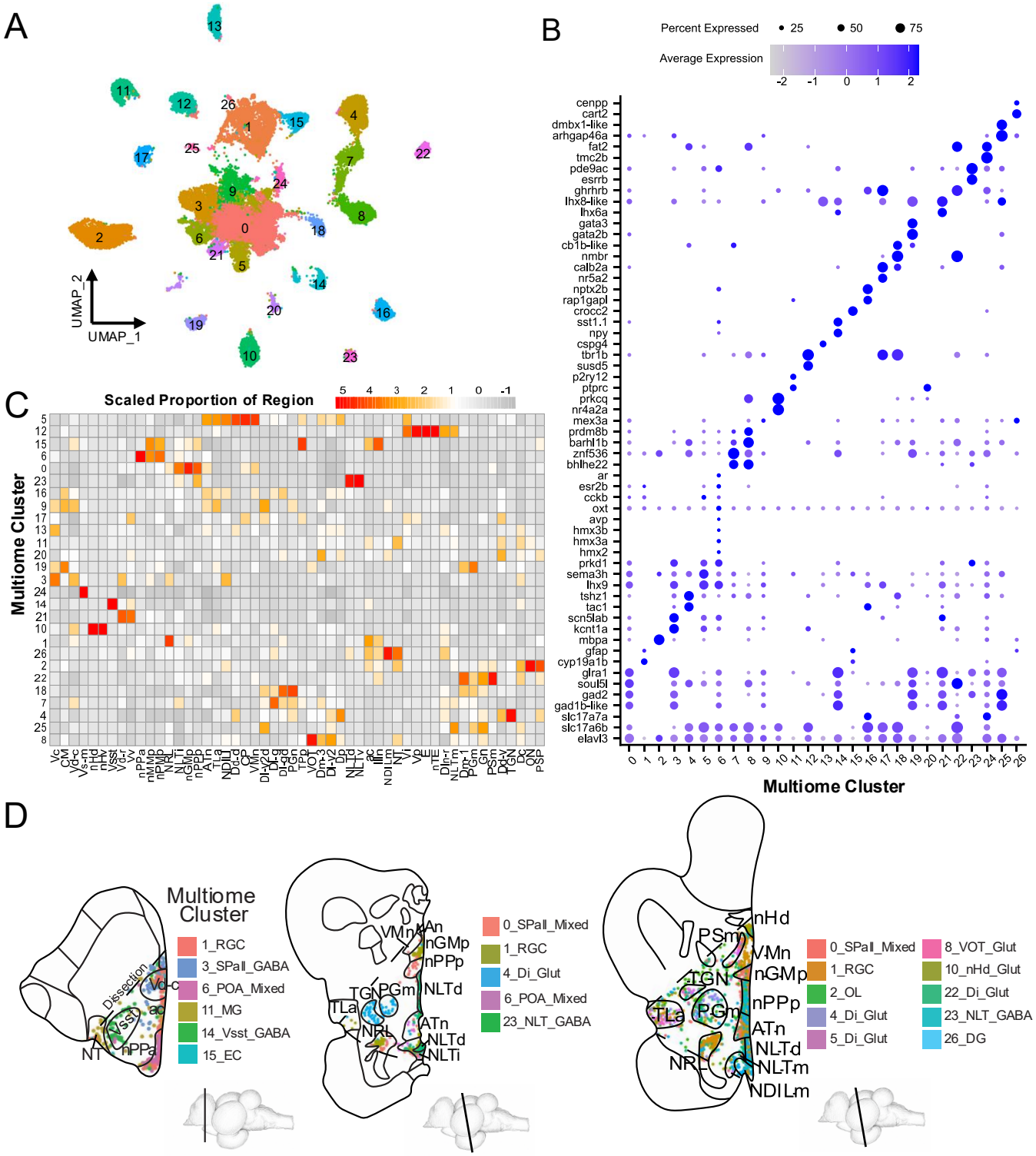
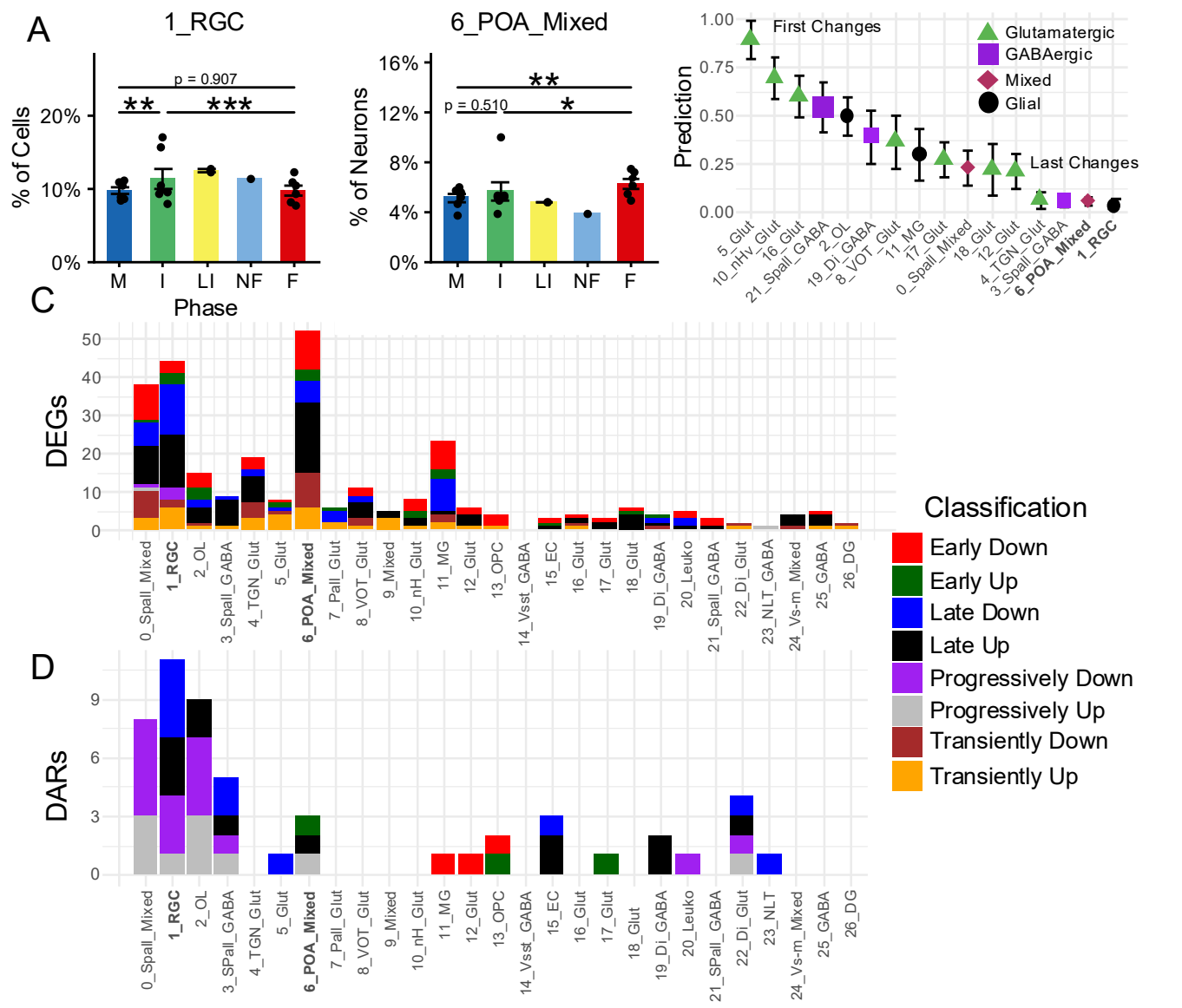
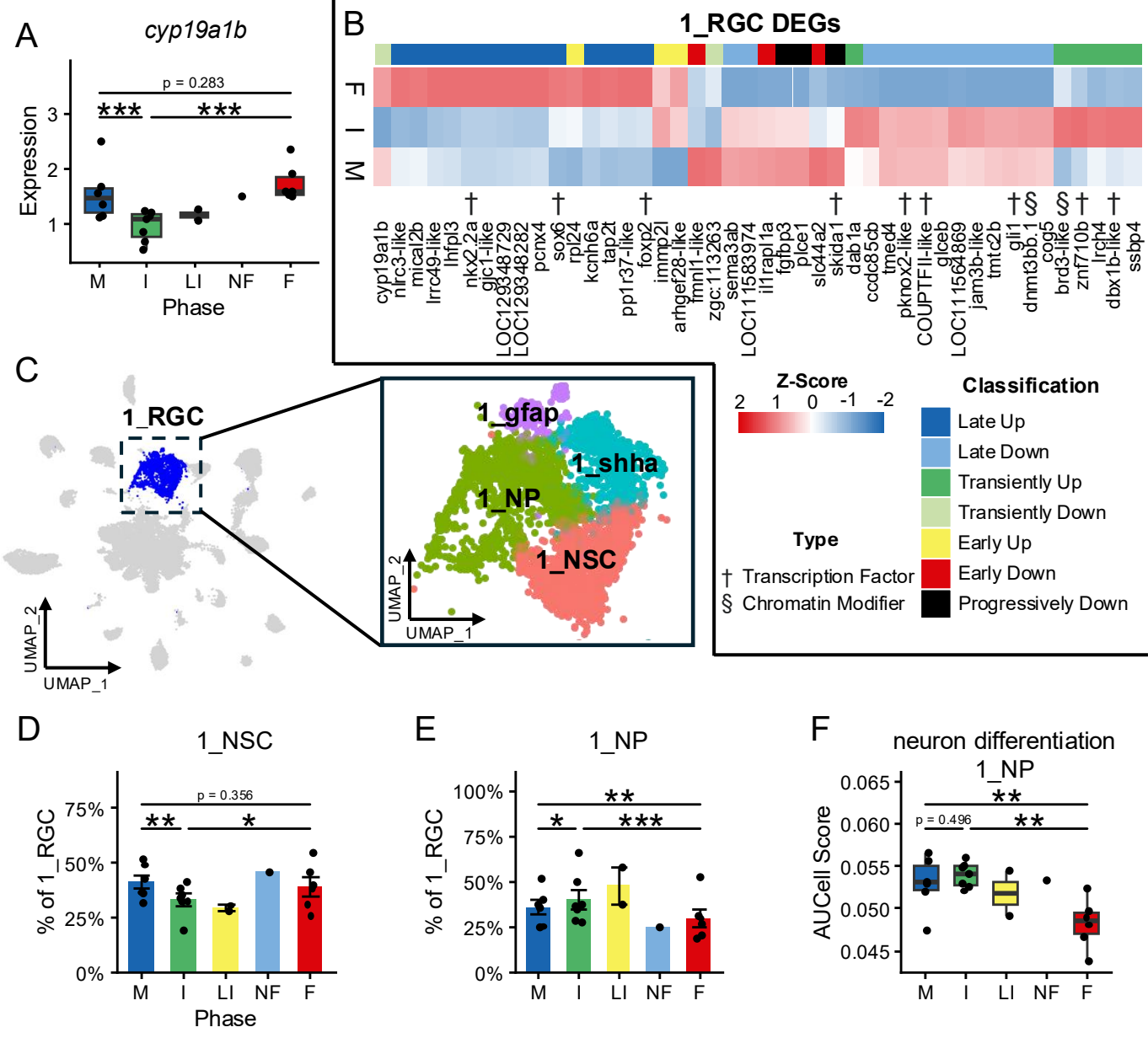
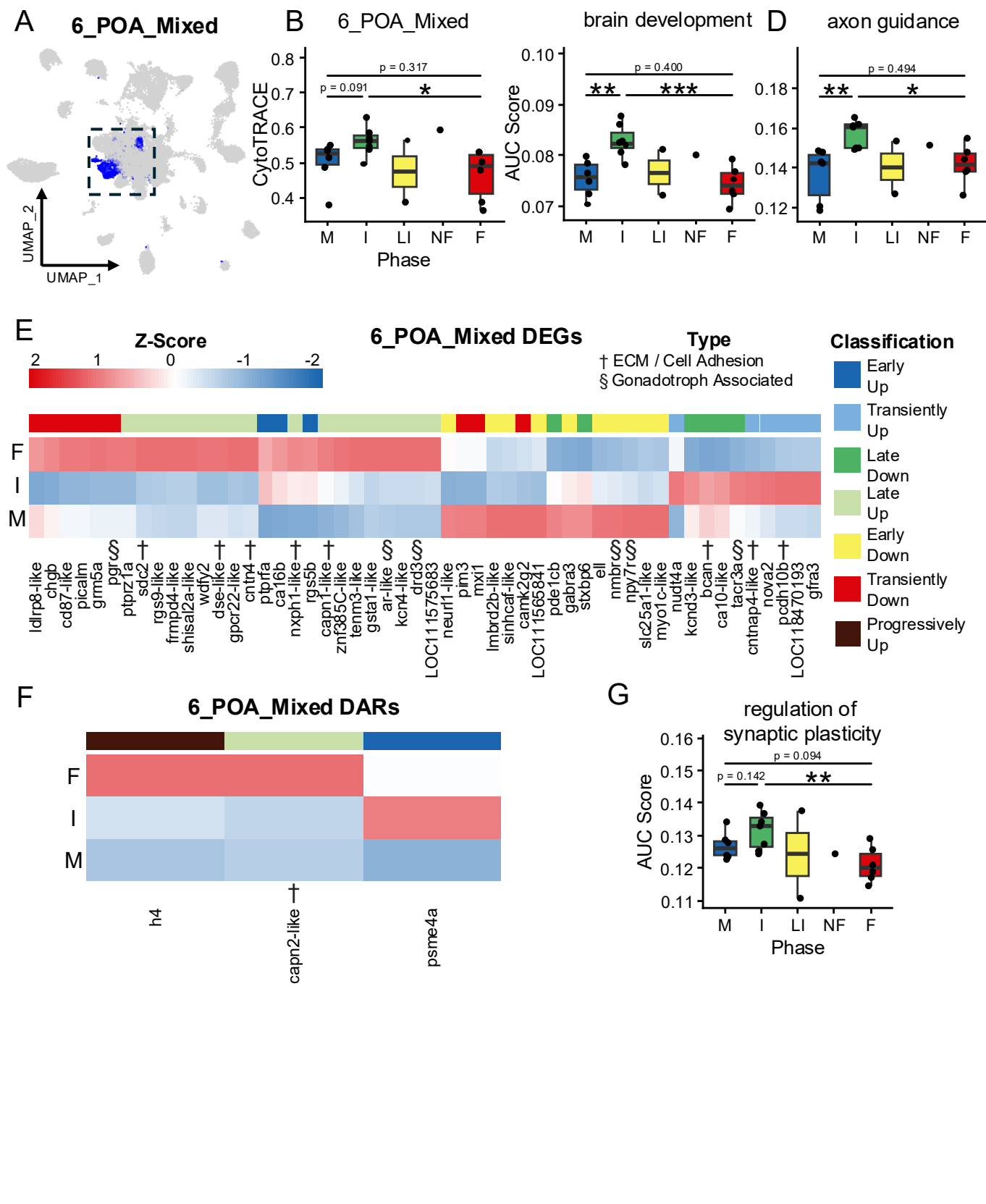
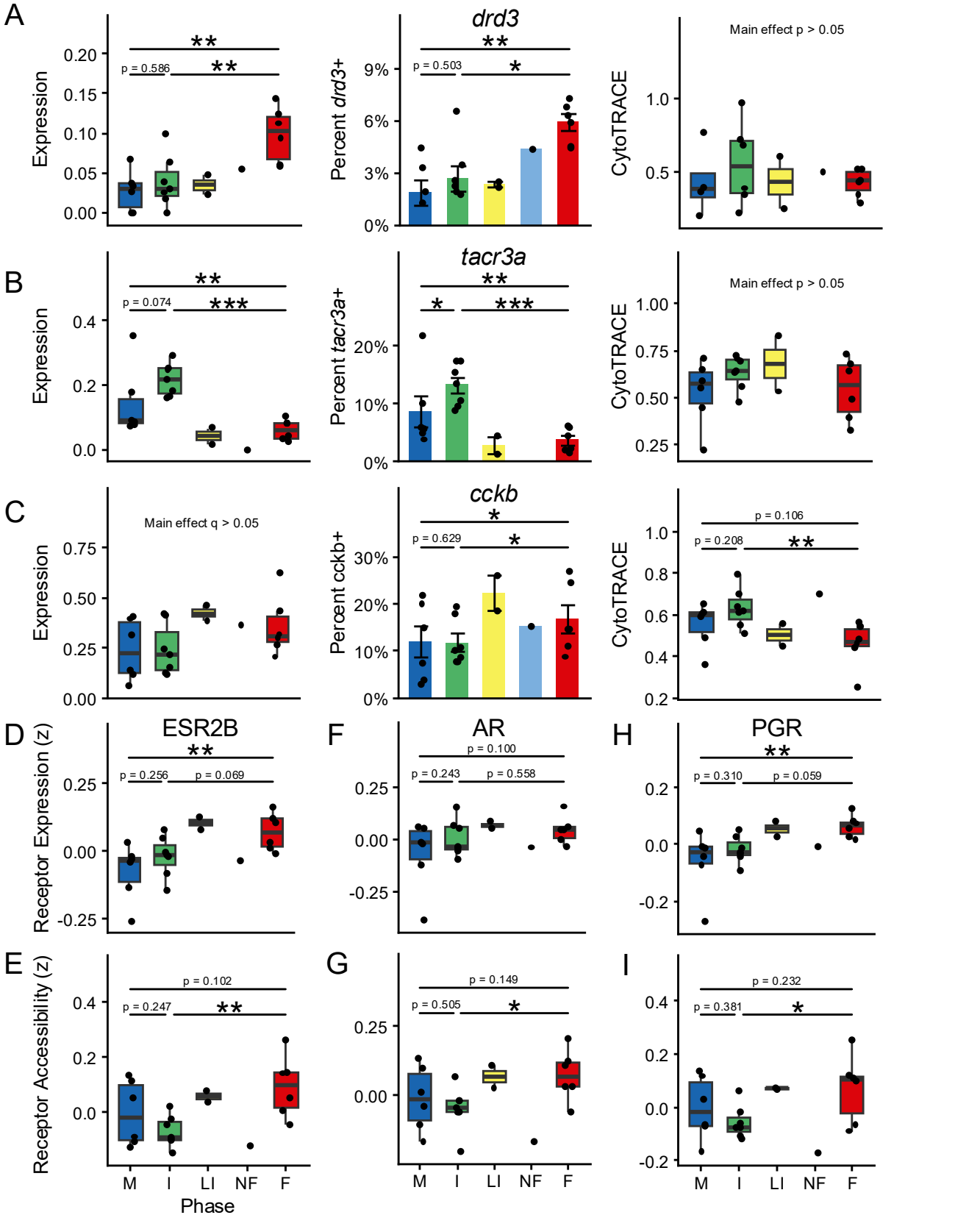

A
Max
Min
Parameter
11KT
E2
Percent Testicular
Vitellogenesis
Body Mass
Neurogenesis
Aggression
Parental Care
I
IP
LI
LIP
Sex Change Progression
B
Dominance
Dominant
Subordinate
Parental Behaviors
Time in nest (s)
Log10 11KT (pg/ml)
2.21x
3.88x
7.64x
11.08x
1.69x
2.14x
1.84x
10.67x
1.85x
1.11x
1.69x
1.35x
F
M
I
IP
LI
LIP
NF
NM
Phase
C
Testicular %
Ovarian %
Corrected log10 volume (px)
M
I
IP
LI
LIP
NM
D
Male
Intermediate
Late Intermediate
Female

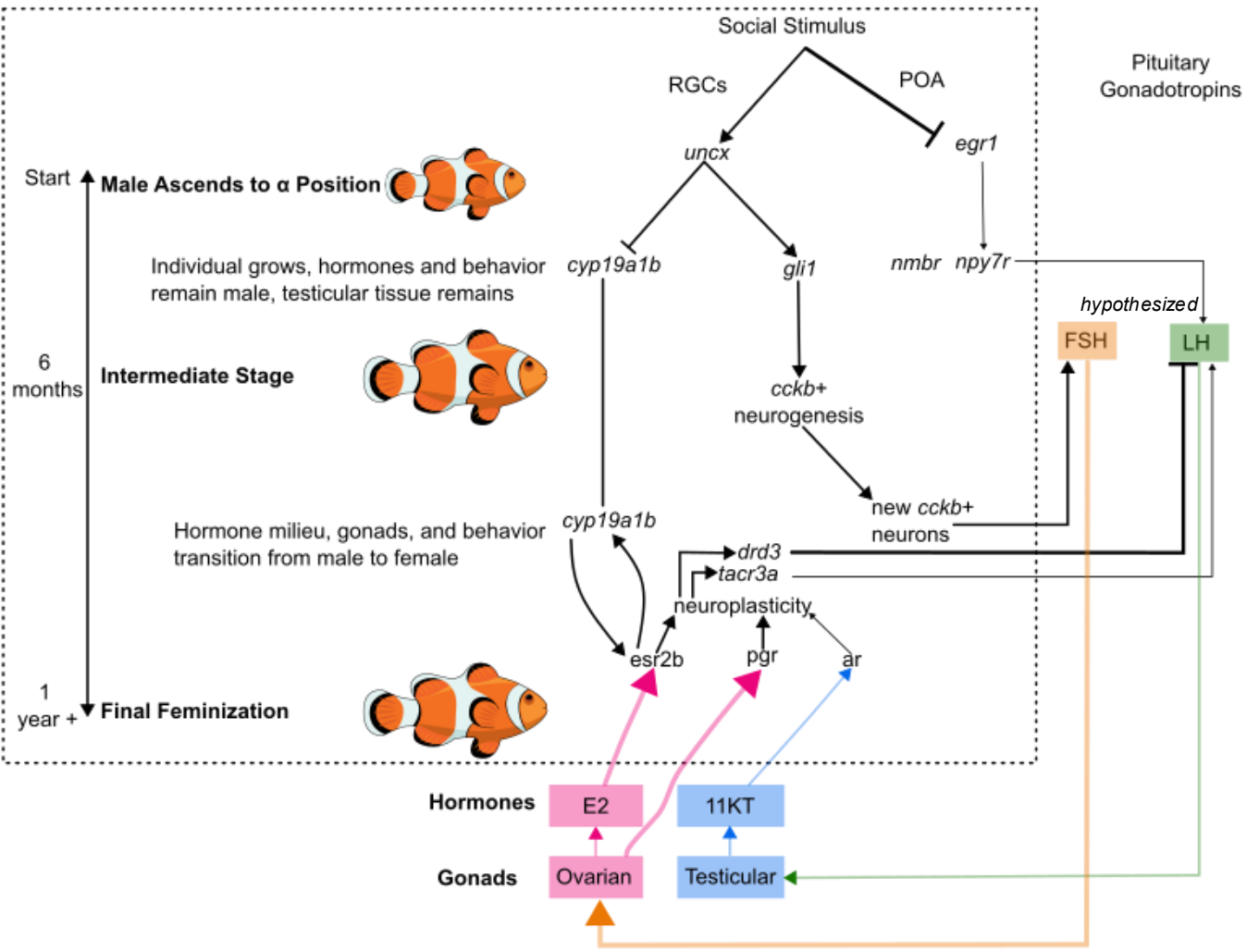

A
UMAP_2
UMAP_1
B
Percent Expressed
25
50
75
Average Expression
-2
-1
0
1
2
Multiome Cluster
C
Scaled Proportion of Region
5
4
3
2
1
0
-1
Multiome Cluster
D
Multiome Cluster
1_RGC
3_SPall_GABA
6_POA_Mixed
11_MG
14_Vsst_GABA
15_EC
Dissection
Vc-c
ac
Vsst
NT
nPPa
VMn
MAn
nGMp
nPPp
TGN
PGm
NLTd
TLa
ATn
NRL
NLTi
0_SPall_Mixed
1_RGC
4_Di_Glut
6_POA_Mixed
23_NLT_GABA
nHd
PSm
VMn
nGMp
TGN
nPPp
TLa
PGm
ATn
NRL
NLTd
NLTm
NDILm
0_SPall_Mixed
1_RGC
2_OL
4_Di_Glut
5_Di_Glut
8_VOT_Glut
10_nHd_Glut
22_Di_Glut
23_NLT_GABA
26_DG

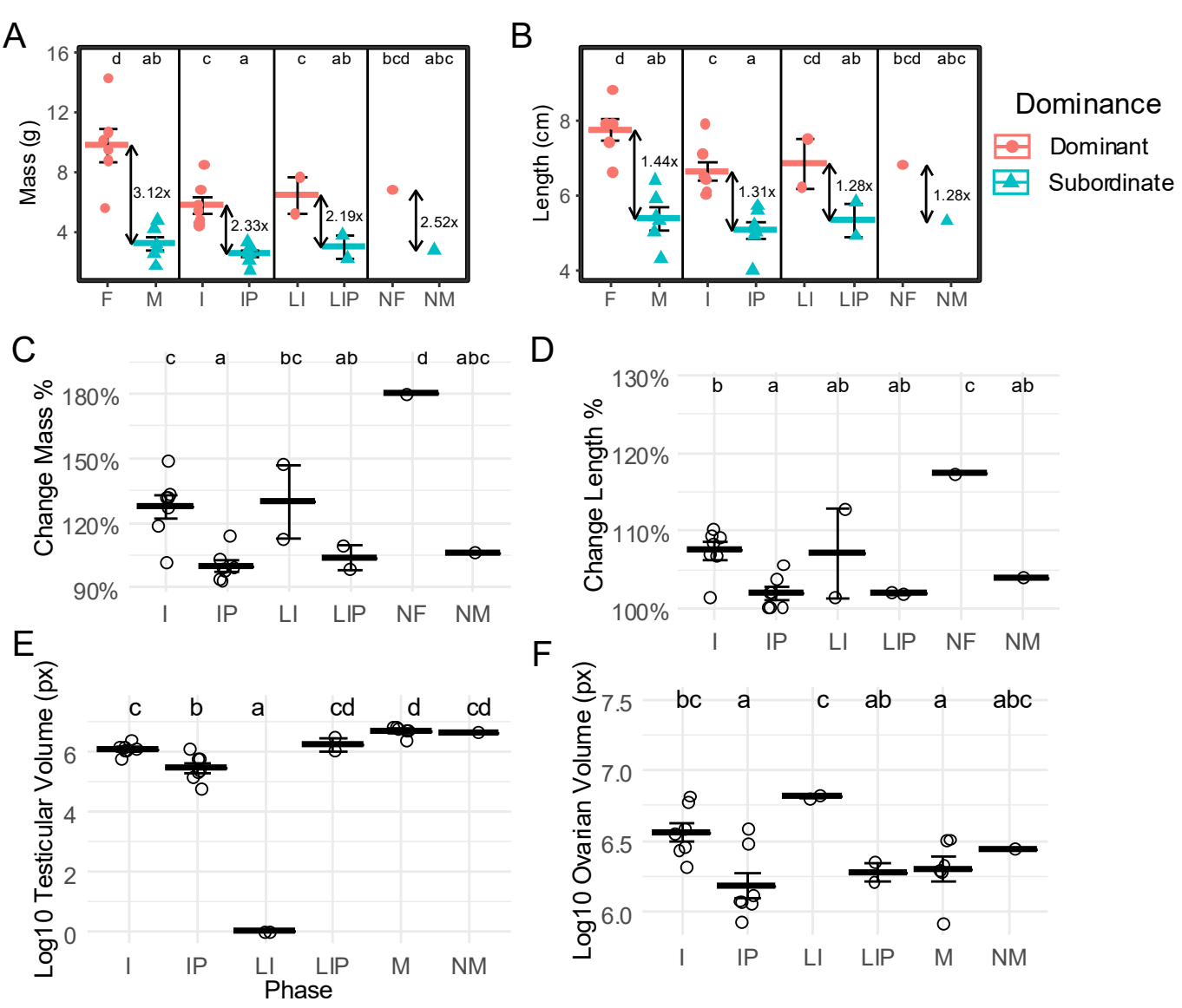

A
1_RGC
6_POA_Mixed
% of Cells
% of Neurons
p = 0.907
p = 0.510
M
I
LI
NF
F
First Changes
Last Changes
Glutamatergic
GABAergic
Mixed
Glial
Prediction
C
Phase
DEGs
Classification
Early Down
Early Up
Late Down
Late Up
Progressively Down
Progressively Up
Transiently Down
Transiently Up
D
DARs


A
cyp19a1b
p = 0.283
***
***
Expression
Phase
M
I
LI
NF
F
B
1_RGC DEGs
F
I
M
cyp19a1b
nfrc3-like
mical2b
lrrc49-like
lhfpl3
nkx2.2a
gjc1-like
LOC129348729
LOC129348282
pcnx4
sox6
rpl24
kcnh6a
tap2t
pp1r37-like
foxp2
immp2l
arhgef28-like
fmnl1-like
zgc:113263
sema3ab
LOC111583974
il1rapl1a
fgfbp3
plce1
slc44a2
skida1
dab1a
ccdc85cb
tmed4
pknox2-like
COUPTFII-like
glceb
LOC111564869
jam3b-like
tmtc2b
gli1
dnmt3bb.1
cog5
brd3-like
znf710b
lrch4
dbx1b-like
ssbp4
C
1_RGC
1_gfap
1_shha
1_NP
1_NSC
UMAP_2
UMAP_1
Z-Score
2 1 0 -1 -2
Classification
Late Up
Late Down
Transiently Up
Transiently Down
Early Up
Early Down
Progressively Down
Type
† Transcription Factor
§ Chromatin Modifier
D
1_NSC
p = 0.356
**
*
% of 1_RGC
Phase
E
1_NP
**
*
***
F
neuron differentiation
1_NP
p = 0.496
**
**
AUCell Score

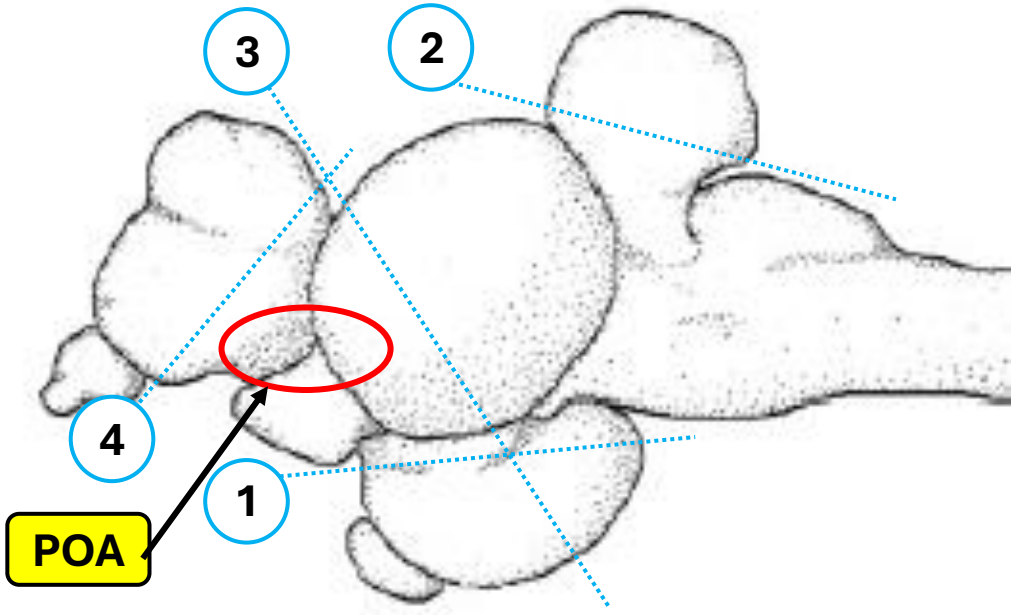

A
6_POA_Mixed
UMAP_2
UMAP_1
B
6_POA_Mixed
CytoTRACE
p = 0.317
p = 0.091
*
M
I
LI
NF
F
Phase
brain development
AUC Score
p = 0.400
**
***
D
axon guidance
p = 0.494
**
*
E
Z-Score
2
1
0
-1
-2
6_POA_Mixed DEGs
Type
† ECM / Cell Adhesion
§ Gonadotroph Associated
Classification
Early Up
Transiently Up
Late Down
Late Up
Early Down
Transiently Down
Progressively Up
F
I
M
ldlrp8-like
chgb
cd87-like
picalm
grm5a
pgrc§
ptprz1a
sdc2†
rgs9-like
frmpd4-like
shisal2a-like
wdfy2
dse-like†
gpcr22-like
cntn4†
ptprfa
ca16b
nxph1-like†
rgs5b
capn1-like†
znf385C-like
tenm3-like
gsta1-like
ar-like§
kcn4-like
drd3§
LOC111575683
neurl1-like
pim3
mxi1
lmbrd2b-like
sinhcaf-like
camk2g2
LOC111565841
pde1cb
gabra3
stxbp6
ell
nmbr§
npy7r§
slc25a1-like
myo1c-like
nudt4a
kcnd3-like†
bcan
ca10-like
tacr3a§
cntnap4-like†
nova2
pcdh10b†
LOC118470193
gfra3
F
6_POA_Mixed DARs
h4
capn2-like†
psme4a
G
regulation of synaptic plasticity
AUC Score
p = 0.094
p = 0.142
**
M
I
LI
NF
F
Phase


A
B
C
D
E
F
G
H
I
Expression
Percent drd3+
Percent tacr3a+
Percent cckb+
CytoTRACE
drd3
tacr3a
cckb
Main effect p > 0.05
Main effect q > 0.05
p = 0.586
p = 0.503
p = 0.074
p = 0.629
p = 0.106
p = 0.208
Receptor Expression (z)
Receptor Accessibility (z)
ESR2B
AR
PGR
p = 0.256
p = 0.069
p = 0.100
p = 0.243
p = 0.558
p = 0.310
p = 0.059
p = 0.102
p = 0.247
p = 0.149
p = 0.505
p = 0.232
p = 0.381
M
I
LI
NF
F
Phase


Social Stimulus
RGCs
POA
Pituitary
Gonadotropins
uncx
egr1
Start
Male Ascends to α Position
Individual grows, hormones and behavior
remain male, testicular tissue remains
cyp19a1b
gli1
nmbr
npy7r
hypothesized
FSH
LH
6
months
Intermediate Stage
cckb+
neurogenesis
new cckb+
neurons
Hormone milieu, gonads, and behavior
transition from male to female
cyp19a1b
drd3
tacr3a
neuroplasticity
esr2b
pgr
ar
1
year +
Final Feminization
Hormones
E2
11KT
Gonads
Ovarian
Testicular

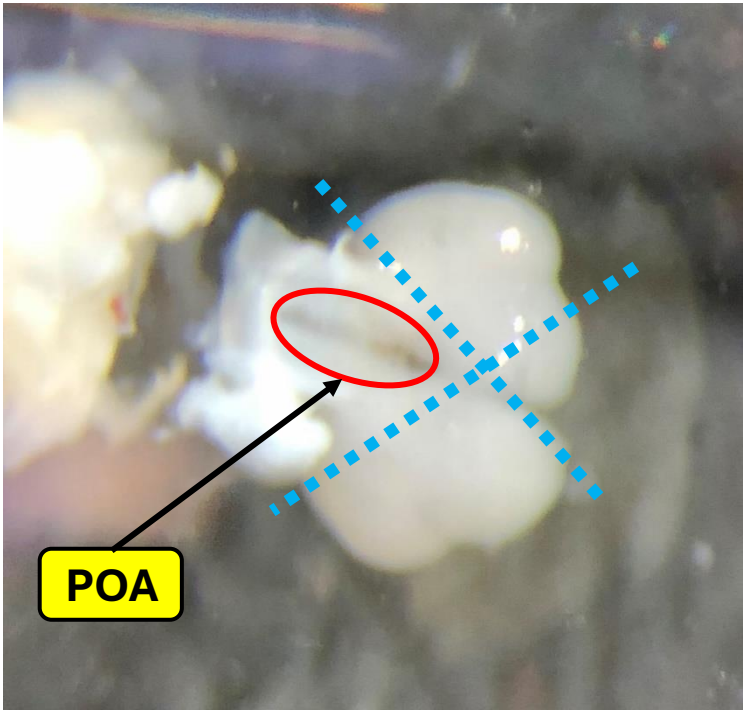

A
B
Mass (g)
Length (cm)
d ab c a c ab bcd abc
d ab c a cd ab bcd abc
3.12x
2.33x
2.19x
2.52x
1.44x
1.31x
1.28x
1.28x
F M I IP LI LIP NF NM
Dominance
Dominant
Subordinate
C
D
Change Mass %
Change Length %
c a bc ab d abc
b a ab ab c ab
E
F
Log10 Testicular Volume (px)
Log10 Ovarian Volume (px)
c b a cd d cd
bc a c ab a abc
I IP LI LIP M NM
Phase

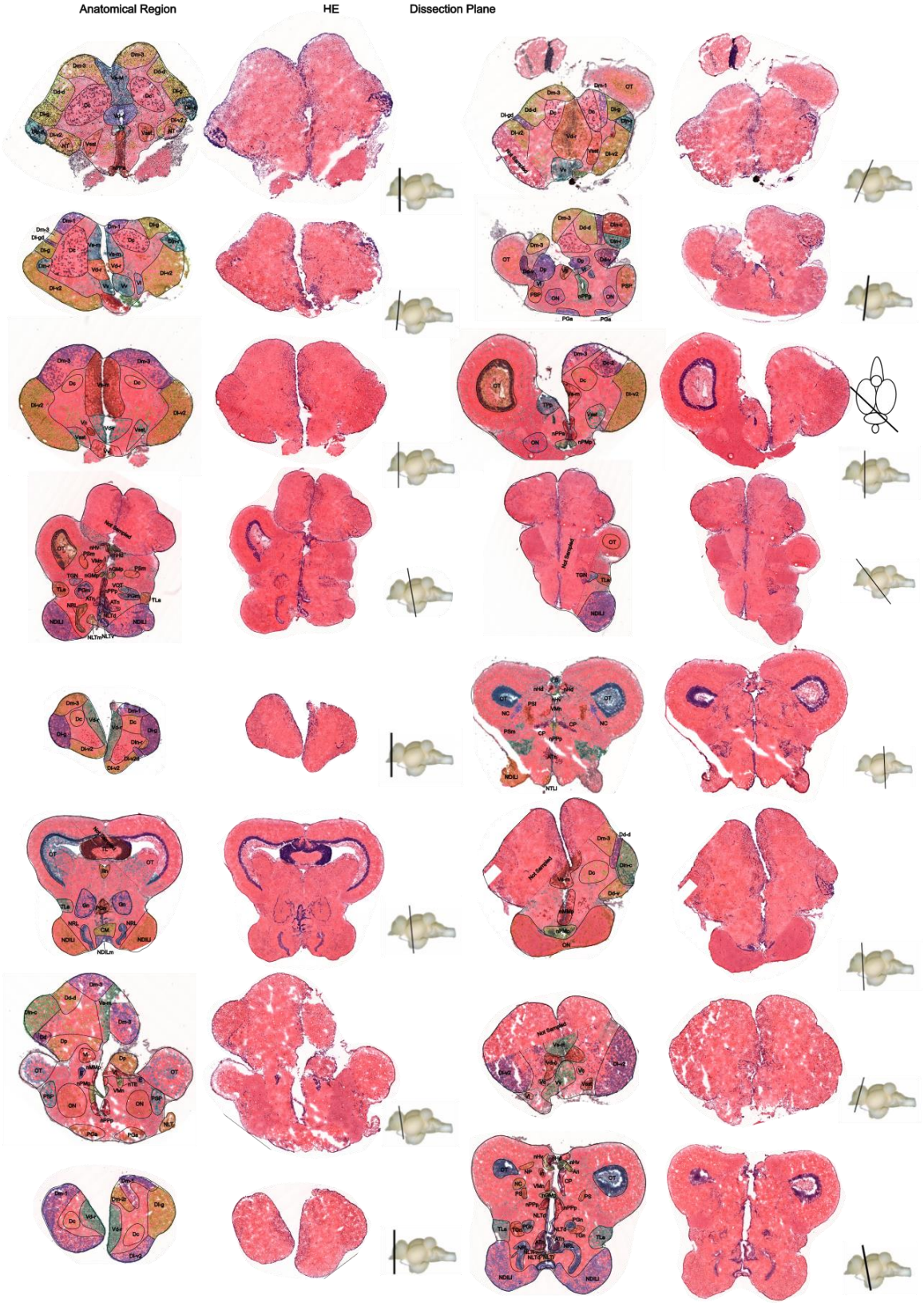
A



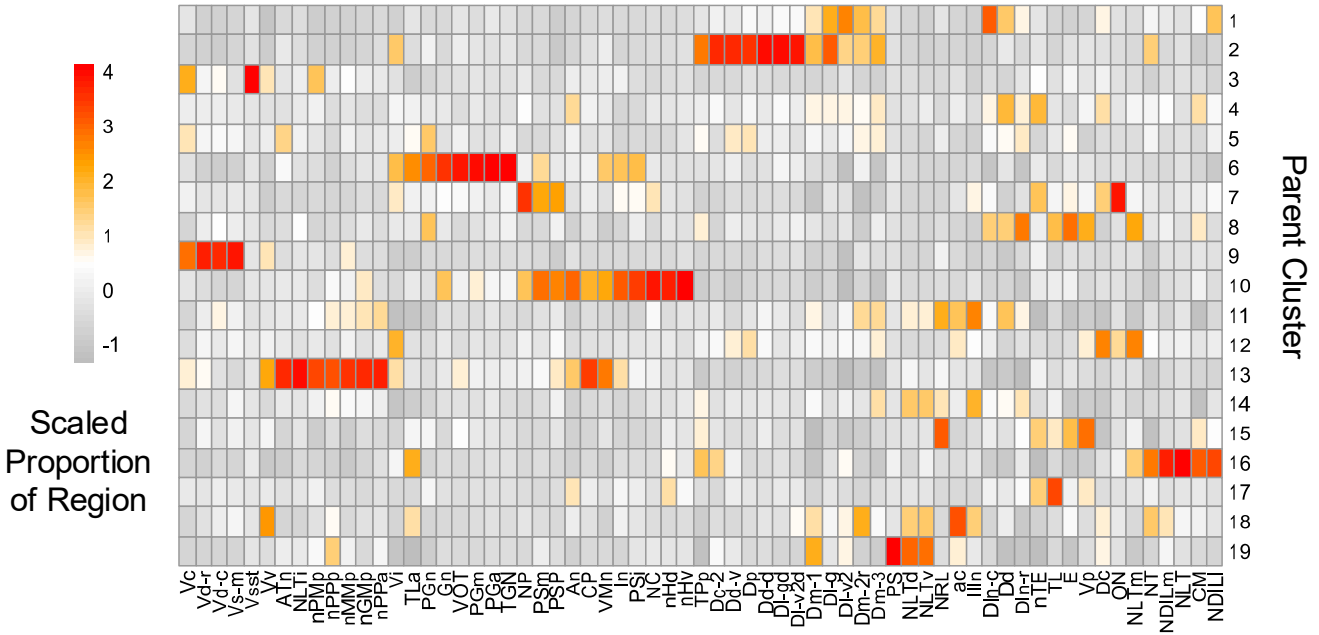
Blue dotted lines represent cuts made

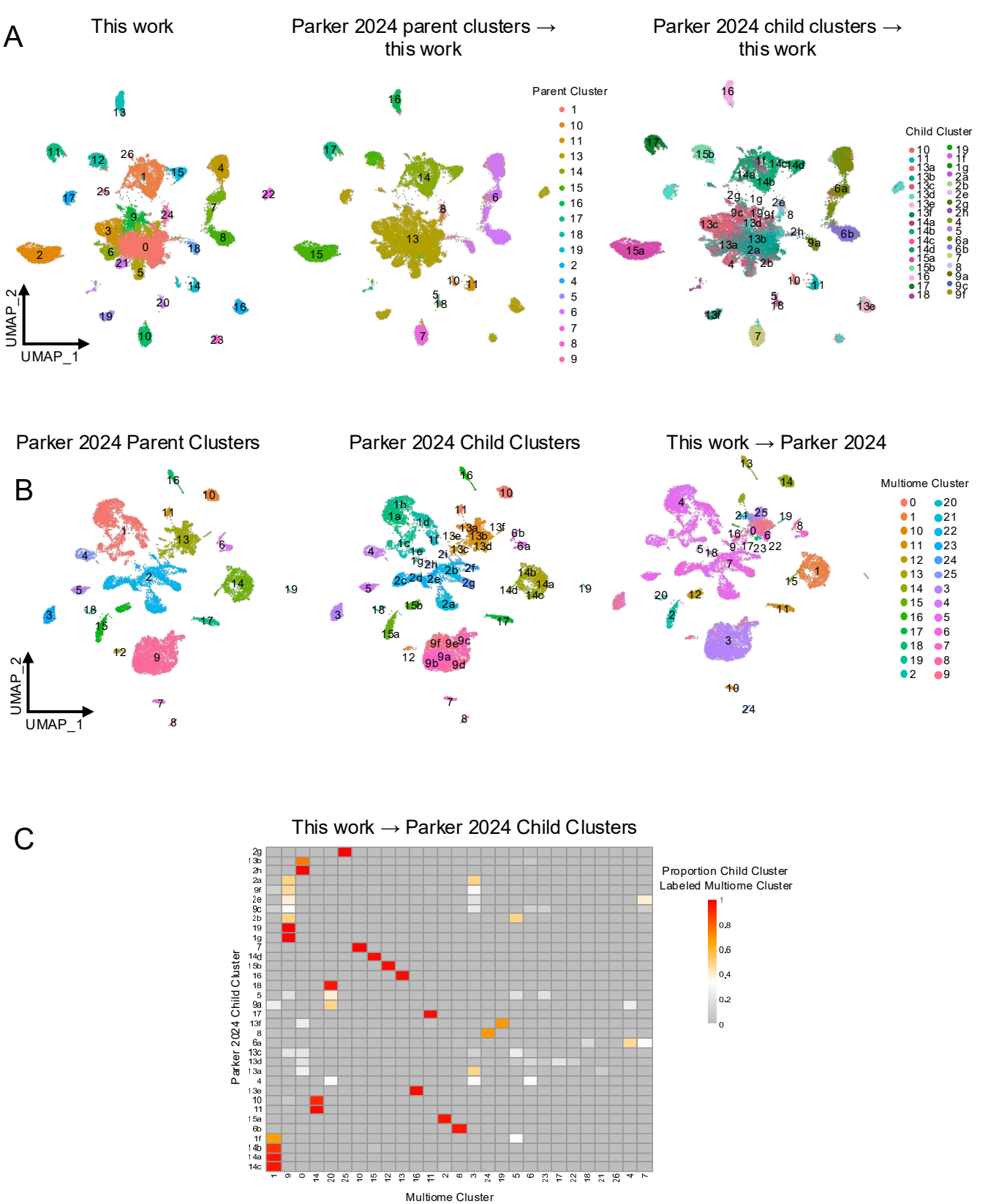
B

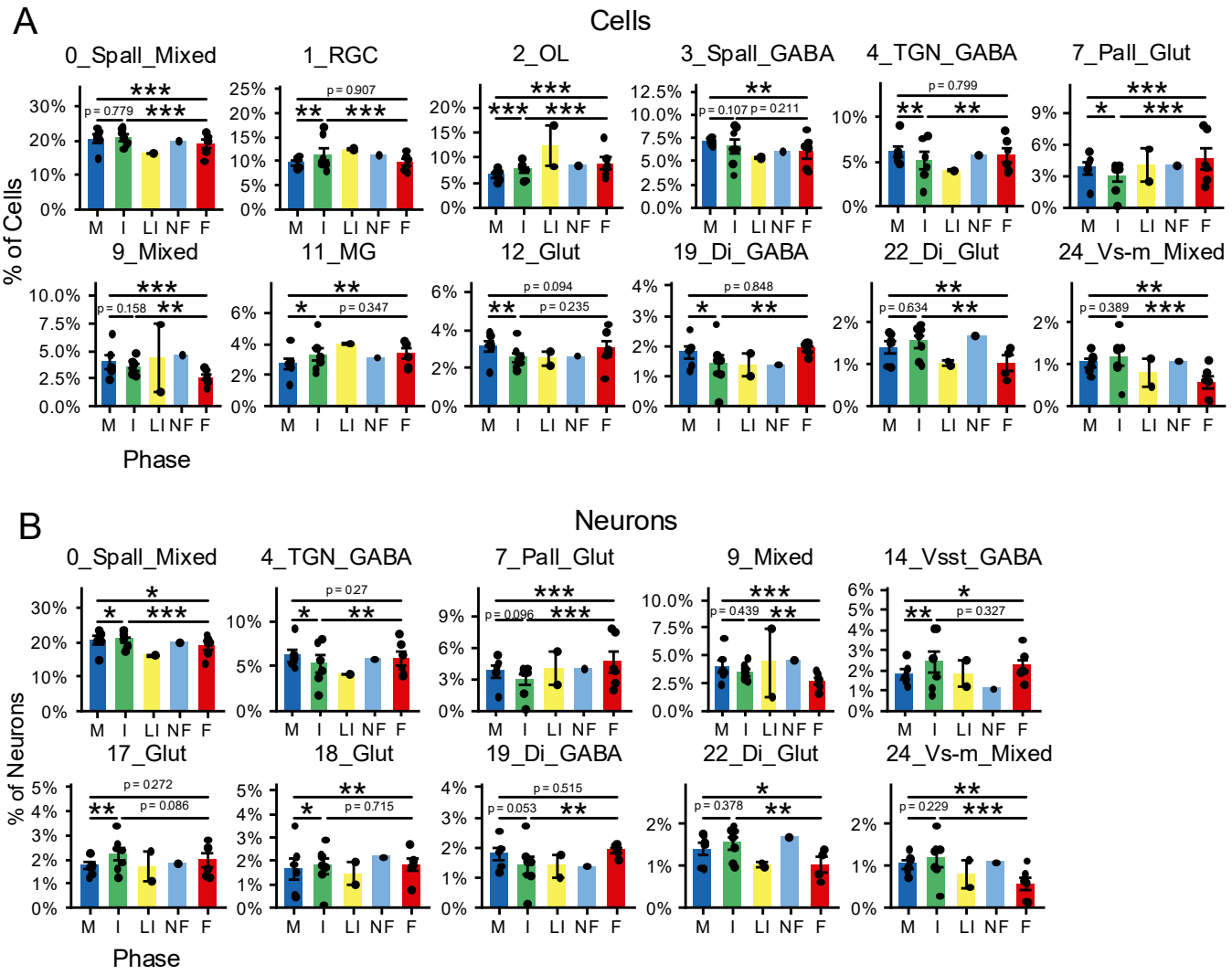


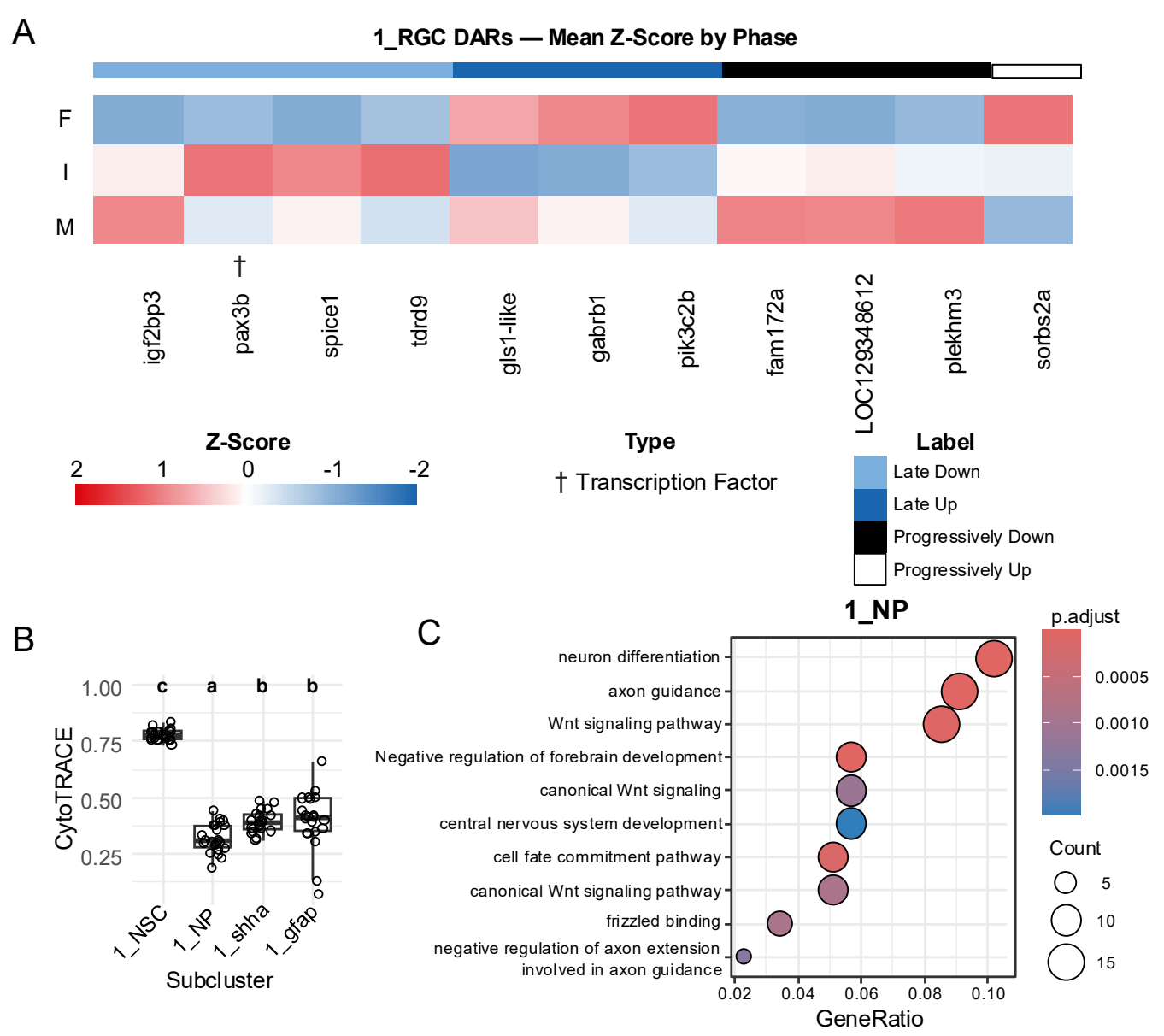

Anatomical Region
HE
Dissection Plane

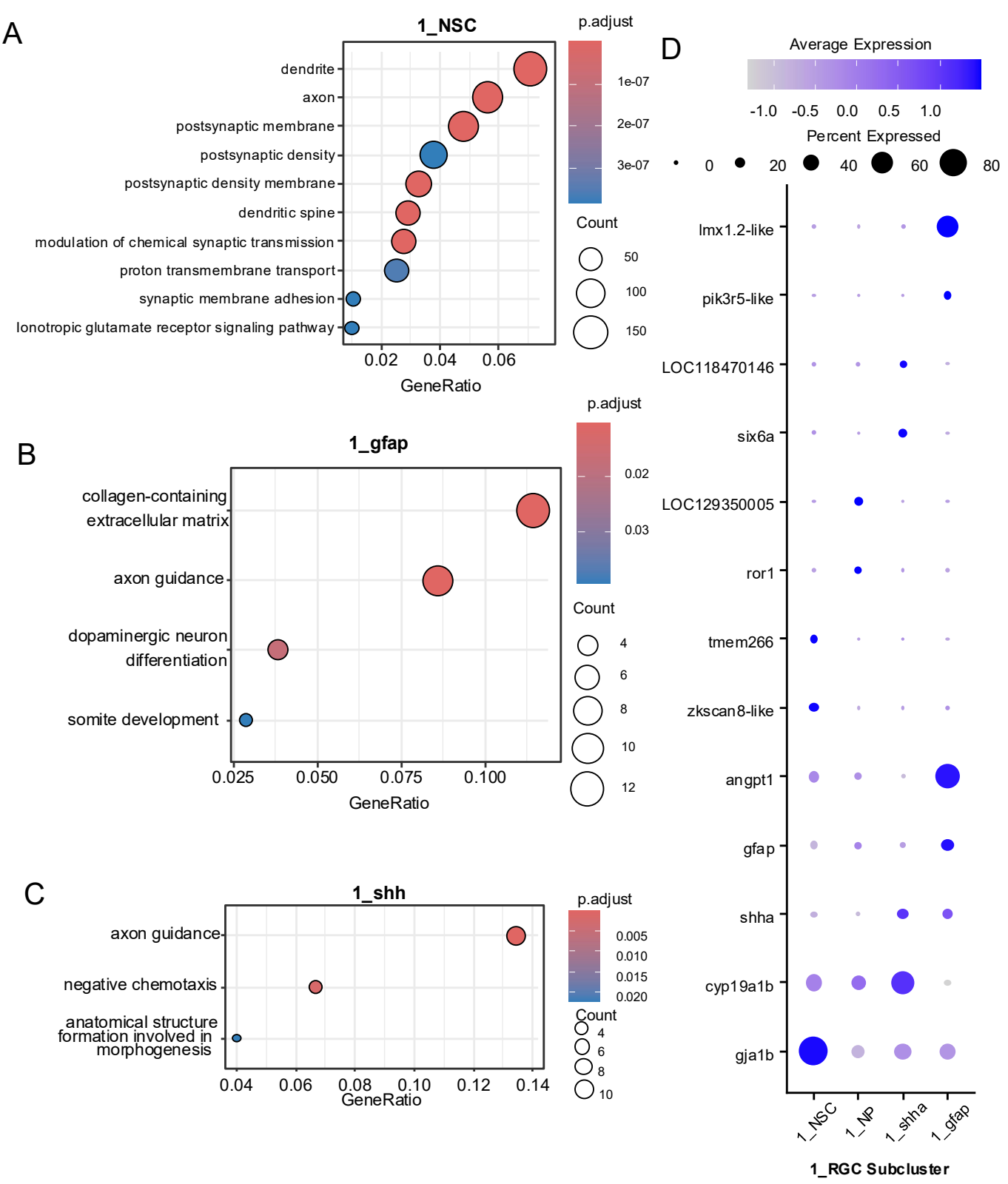
Parker 2024 Parent Clusters

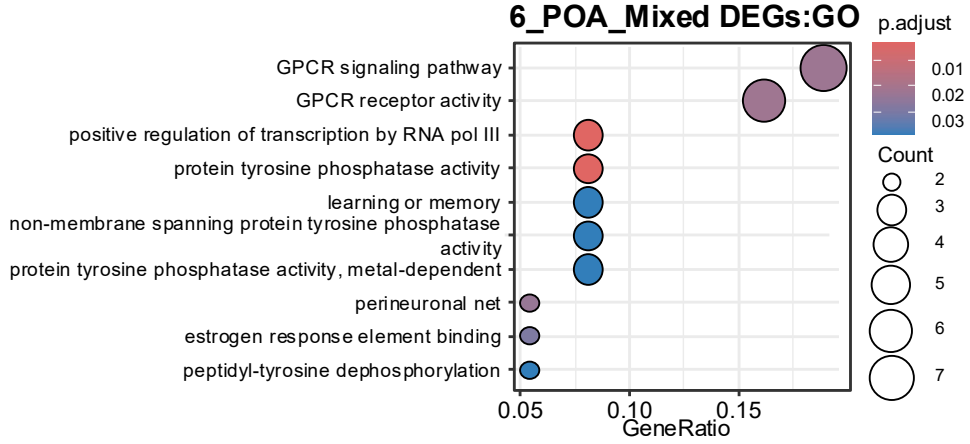


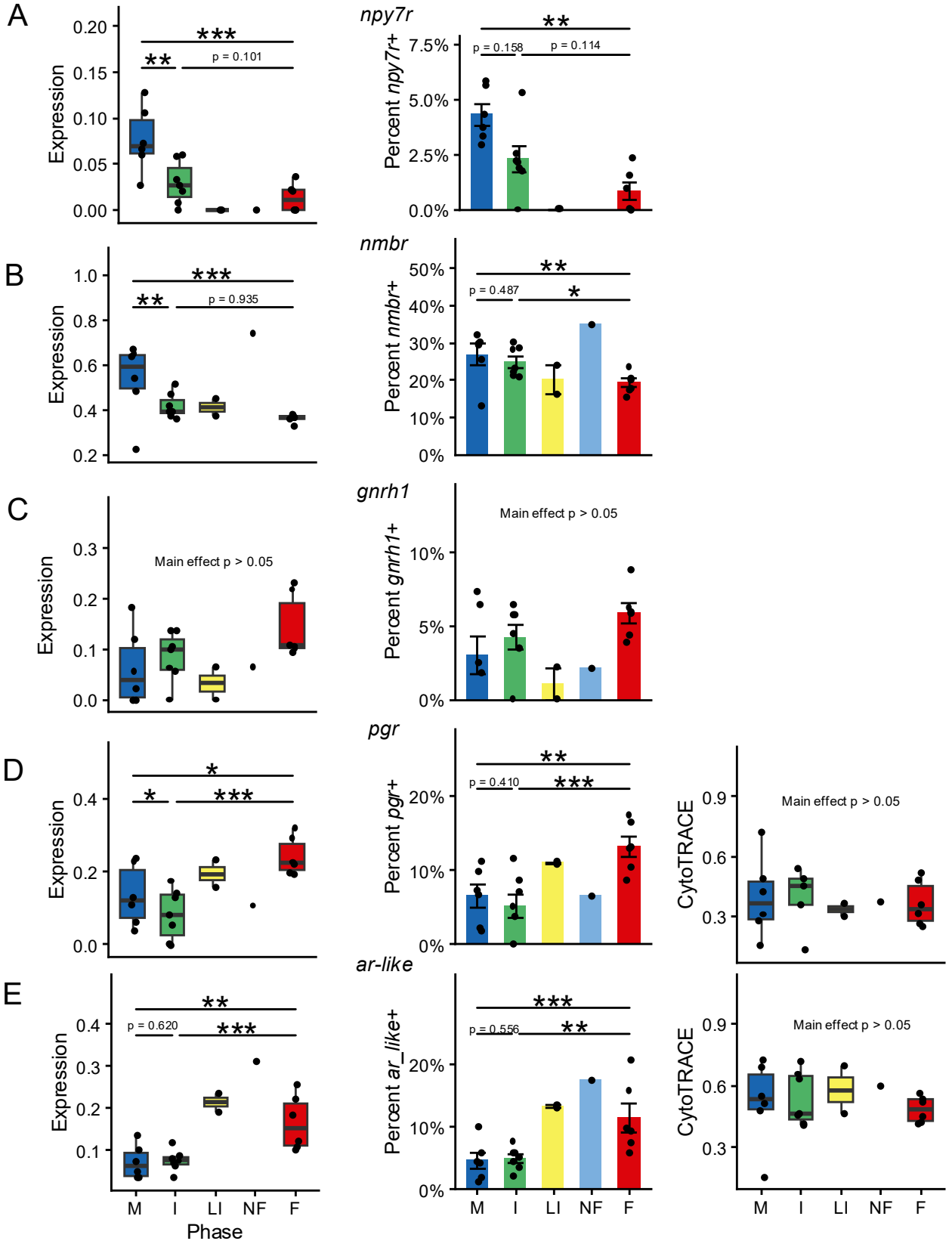

A
This work
Parker 2024 parent clusters → this work
Parker 2024 child clusters → this work
Parent Cluster
Child Cluster
UMAP_2
UMAP_1
B
Parker 2024 Parent Clusters
Parker 2024 Child Clusters
This work → Parker 2024
Multiome Cluster
C
This work → Parker 2024 Child Clusters
Parker 2024 Child Cluster
Multiome Cluster
Proportion Child Cluster Labeled Multiome Cluster

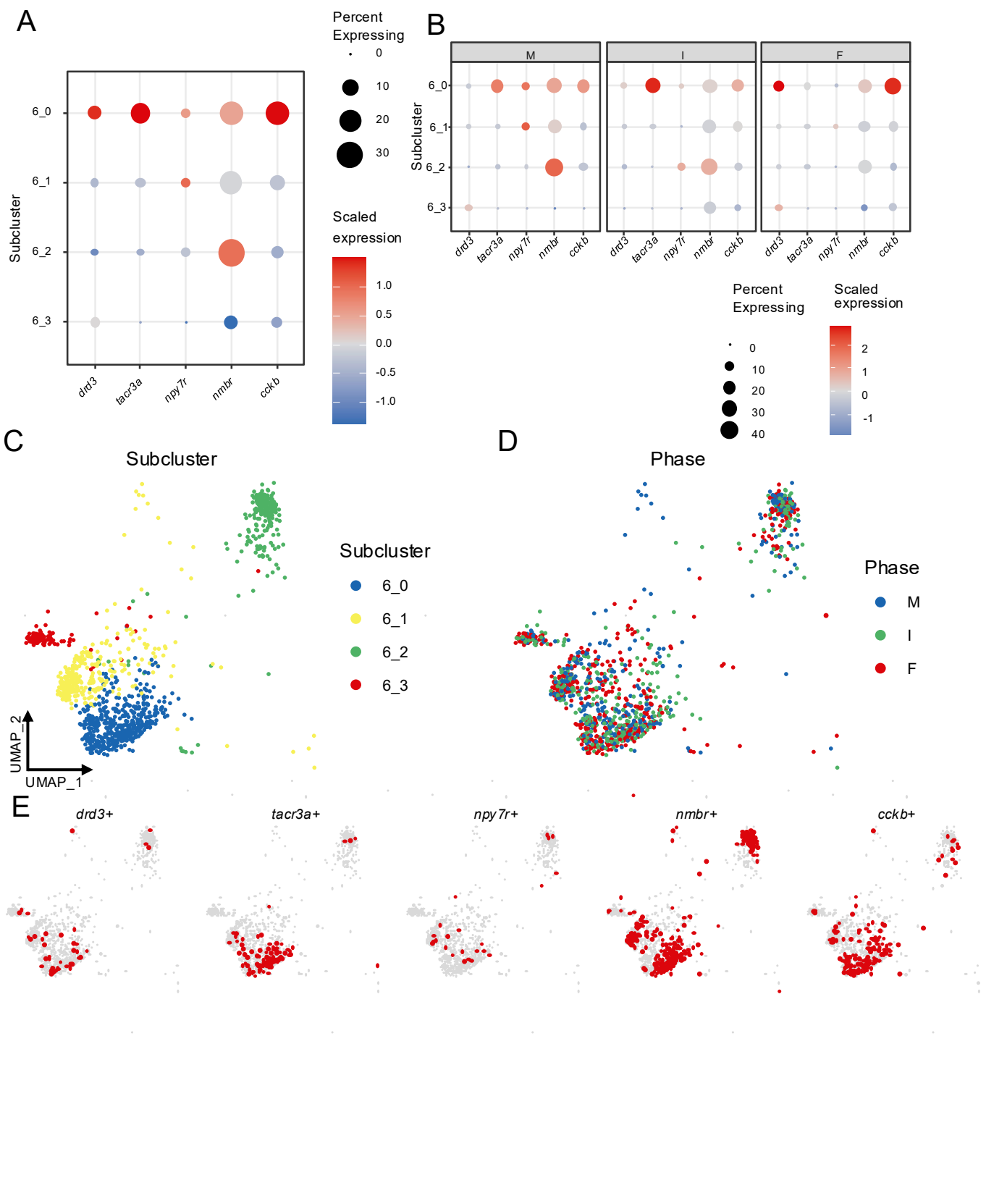

A
Cells
0_Spall_Mixed
1_RGC
2_OL
3_Spall_GABA
4_TGN_GABA
7_Pall_Glut
9_Mixed
11_MG
12_Glut
19_Di_GABA
22_Di_Glut
24_Vs-m_Mixed
% of Cells
Phase
M I LI NF F
B
Neurons
0_Spall_Mixed
4_TGN_GABA
7_Pall_Glut
9_Mixed
14_Vsst_GABA
17_Glut
18_Glut
19_Di_GABA
22_Di_Glut
24_Vs-m_Mixed
% of Neurons
Phase

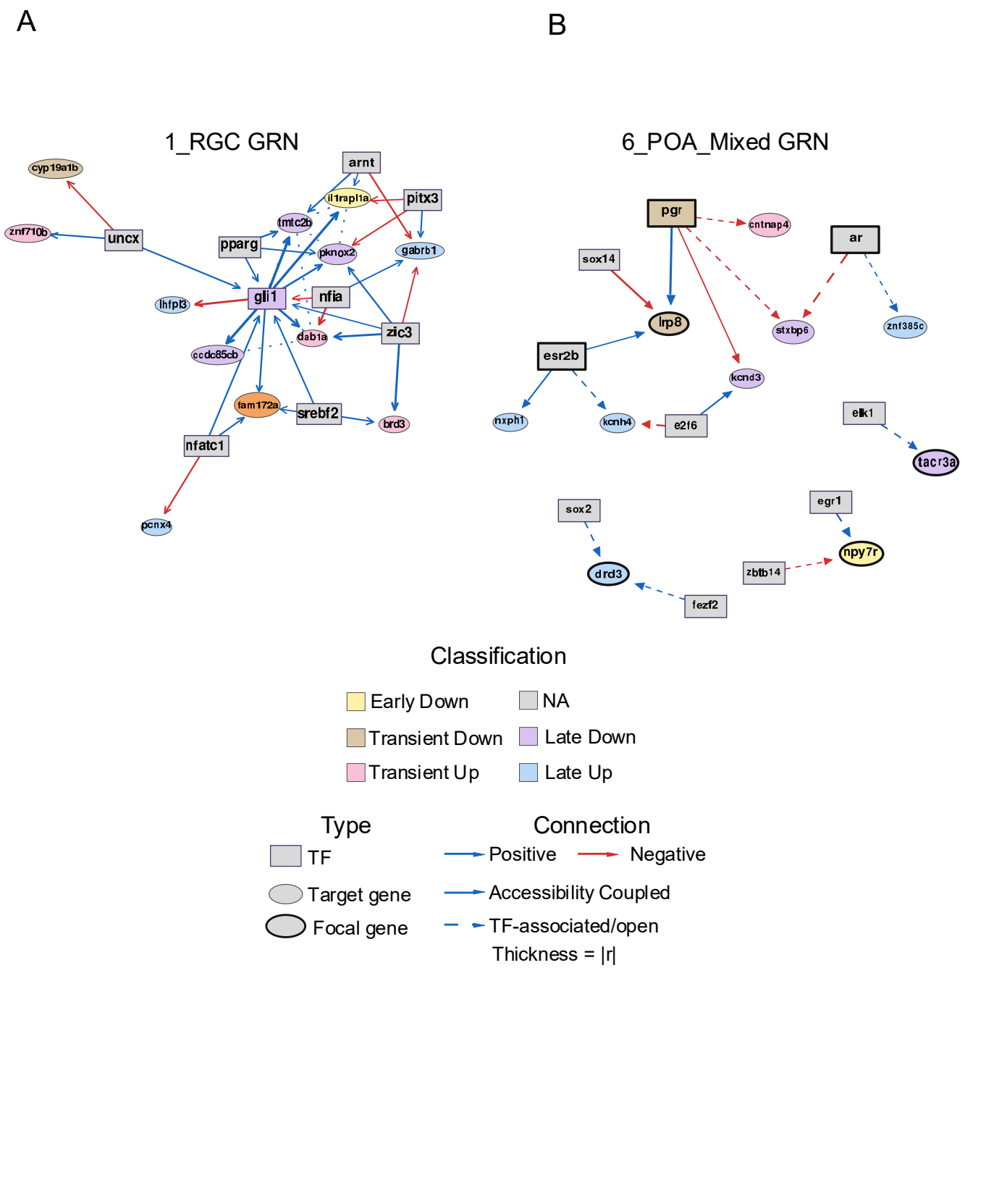

A
1_RGC DARs — Mean Z-Score by Phase
F
I
M
†
igf2bp3
pax3b
spice1
tdrd9
gls1-like
gabrb1
pik3c2b
fam172a
LOC129348612
plekhm3
sorbs2a
Z-Score
2
1
0
-1
-2
Type
† Transcription Factor
Label
Late Down
Late Up
Progressively Down
Progressively Up
B
1.00
0.75
0.50
0.25
CytoTRACE
c
a
b
b
1_NSC
1_NP
1_shha
1_gfap
Subcluster
C
1_NP
neuron differentiation
axon guidance
Wnt signaling pathway
Negative regulation of forebrain development
canonical Wnt signaling
central nervous system development
cell fate commitment pathway
canonical Wnt signaling pathway
frizzled binding
negative regulation of axon extension involved in axon guidance
0.02
0.04
0.06
0.08
0.10
GeneRatio
p.adjust
0.0005
0.0010
0.0015
Count
5
10
15


A
1_NSC
dendrite
axon
postsynaptic membrane
postsynaptic density
postsynaptic density membrane
dendritic spine
modulation of chemical synaptic transmission
proton transmembrane transport
synaptic membrane adhesion
Ionotropic glutamate receptor signaling pathway
0.02
0.04
0.06
GeneRatio
p.adjust
1e-07
2e-07
3e-07
Count
50
100
150
B
1_gfap
collagen-containing extracellular matrix
axon guidance
dopaminergic neuron differentiation
somite development
0.025
0.050
0.075
0.100
GeneRatio
p.adjust
0.02
0.03
Count
4
6
8
10
12
C
1_shh
axon guidance
negative chemotaxis
anatomical structure formation involved in morphogenesis
0.04
0.06
0.08
0.10
0.12
0.14
GeneRatio
p.adjust
0.005
0.010
0.015
0.020
Count
4
6
8
10
D
Average Expression
-1.0
-0.5
0.0
0.5
1.0
Percent Expressed
0
20
40
60
80
lmx1.2-like
pik3r5-like
LOC118470146
six6a
LOC129350005
ror1
tmem266
zkscan8-like
angpt1
gfap
shha
cyp19a1b
gja1b
1_NSC
1_NP
1_shha
1_gfap
1_RGC Subcluster


6_POA_Mixed DEGs:GO
GPCR signaling pathway
GPCR receptor activity
positive regulation of transcription by RNA pol III
protein tyrosine phosphatase activity
learning or memory
non-membrane spanning protein tyrosine phosphatase activity
protein tyrosine phosphatase activity, metal-dependent
perineuronal net
estrogen response element binding
peptidyl-tyrosine dephosphorylation
0.05
0.10
0.15
GeneRatio
p.adjust
0.01
0.02
0.03
Count
2
3
4
5
6
7


A
B
C
D
E
Expression
Percent npy7r+
Percent nmbr+
Percent gnrh1+
Percent pgr+
Percent ar_like+
CytoTRACE
npy7r
nmbr
gnrh1
pgr
ar-like
p = 0.101
p = 0.158
p = 0.114
p = 0.935
p = 0.487
Main effect p > 0.05
p = 0.410
p = 0.620
p = 0.556
M
I
LI
NF
F
Phase


A
Percent Expressing
0
10
20
30
Scaled expression
1.0
0.5
0.0
-0.5
-1.0
Subcluster
6_0
6_1
6_2
6_3
drd3
tacr3a
npy7r
nmbr
cckb
B
M
I
F
Percent Expressing
0
10
20
30
40
Scaled expression
2
1
0
-1
C
Subcluster
Subcluster
6_0
6_1
6_2
6_3
UMAP_2
UMAP_1
D
Phase
Phase
M
I
F
E
drd3+
tacr3a+
npy7r+
nmbr+
cckb+


A
B
1_RGC GRN
6_POA_Mixed GRN
arnt
cyp19a1b
il1rapl1a
pitx3
tmtc2b
znf710b
uncx
pparg
pknox2
gabrb1
lhfpl3
gli1
nfia
dab1a
zic3
ccdc85cb
fam172a
srebf2
brd3
nfatc1
pcnx4
pgr
cntnap4
ar
sox14
lrp8
stxbp6
znf385c
esr2b
kcnd3
nxph1
kcnh4
e2f6
elk1
tacr3a
sox2
egr1
npy7r
drd3
zbtb14
fezf2
Classification
Early Down
NA
Transient Down
Late Down
Transient Up
Late Up
Type
TF
Target gene
Focal gene
Connection
Positive
Negative
Accessibility Coupled
TF-associated/open
Thickness = |r|